

Employee & Sales Analytics

Data Analysis Project Report

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1. Executive Summary

This project analyses a synthetic dataset of 250 employees across 7 departments and 5,000 sales transactions spanning 2022–2025. Analysis is performed end-to-end using **SQL**, **Excel**, **Power BI**, **Python (Pandas & NumPy)** to demonstrate a full data-analyst toolchain.

2. Datasets

File	Rows	Description
employees.csv	250	HR master — dept, salary, rating, status
sales.csv	5000	5,000 orders 2022–2025
products.csv	26	5 categories, 26 SKUs
customers.csv	400	4 segments, 5 regions
departments.csv	7	7 departments

3. Tools & Techniques

Tool	Purpose
SQL	Schema design, ETL, business queries, window functions (YoY)
Excel	Pivot tables, charts (bar, pie, line), formatted KPI sheets
Pandas	Group-by, joins, RFM segmentation, correlation
NumPy	Vectorised binning (np.select), random data generation
Power BI	Interactive dashboards, DAX measures, slicers
Plotly	HTML dashboards (Power BI alternative preview)

4. Key Findings — Employee

Department	Total	Exited	Attrition %
Marketing	33.0	9.0	27.27
Finance	34.0	8.0	23.53
Human Resources	41.0	9.0	21.95
Customer Support	34.0	6.0	17.65
Sales	37.0	6.0	16.22
Operations	37.0	5.0	13.51
Engineering	34.0	4.0	11.76

Overall attrition rate: 18.80%

Average salary: \$62,379

Average performance rating: 3.47/5

5. Key Findings — Sales

Year	Revenue (\$)	Profit (\$)	YoY %
2022	1,241,852	430,250	—
2023	1,697,585	599,962	36.70%
2024	2,190,164	757,956	29.02%
2025	1,927,686	684,725	-11.98%

Total revenue (2022–2025): \$7,057,287

Total profit: \$2,472,893

Overall margin: 35.04%

6. Business Recommendations

1. Prioritise retention programs in the two highest-attrition departments.
2. Realign sales incentives — reward reps with the best revenue-to-salary ratio.
3. Double-down on the top-selling product category — it accounts for the largest share of revenue & profit.
4. Investigate Q4 sales spike drivers and replicate the tactics in other quarters.
5. Diversify regional presence — over-reliance on any single region increases risk.

7. Deliverables in this Package

- **01_datasets/** — 5 CSV files (departments, employees, products, customers, sales)
- **02_sql/** — schema, load, and 25+ analysis queries
- **03_excel/** — 2 workbooks with pivot tables & native charts
- **04_python_notebooks/** — 4 Jupyter notebooks (Pandas + NumPy)
- **05_powerbi/** — step-by-step build guide + DAX formulas
- **06_reports/** — this PDF
- **07_html_dashboards/** — interactive Plotly dashboards (Power BI preview)